**Author:** Pankaj Yadav
**Date:** 04/12/2026
**Description:**
This assignment is addition to existing milestone 1 & additional prompts.

**Explanation**

The goal of this project is to develop a generative AI career coaching expert. This AI will provide personalized, actionable career advice, help users identify and articulate their strengths, suggest skill development paths, and guide them through job search strategies tailored to their target roles and industries. The model will leverage industry-specific knowledge and hiring trends to deliver high-impact, context-aware coaching.

In [31]:
```
import ipywidgets as widgets
import openai
import docx
import os
```

Provided location for my resume & a job description.

In [32]:
```
api_key = os.getenv("OPENAI_API_KEY")
file_path = "Resume/Pankaj_Yadav.docx"
job_path = "Resume/JD_Equifax.txt"
```

Created this function initially to just read my resume but reusing it for job description too.

In [33]:
```
def read_resume(file_path):
    if file_path.endswith('.docx'):
        doc = docx.Document(file_path)
        return '\n'.join([para.text for para in doc.paragraphs])
    elif file_path.endswith('.txt'):
        with open(file_path, 'r') as f:
            return f.read()
    else:
        raise ValueError("Unsupported file format. Use .docx or .txt")

resume_text = read_resume(file_path)
print(resume_text[:100])  # Print first 100 characters as a preview
```

```
PANKAJ YADAV | +1-210-294-1837 | pankaj.y@icloud.com | linkedin.com/in/pankajkyadav

SUMMARY
Highly
```

In [34]:
```python
job_description = read_resume(job_path)
print(job_description[:100])
```

```
We are seeking an experienced and driven Business Intelligence expert to architect and scale an indu
```

Lets define a function based on OpenAI's API documentation (https://developers.openai.com/api/docs/guides/prompt-engineering?lang=python) for responses. I will use this function to fetch the results and print them. this will make the code little sleek.

In [35]:
```python
def openai_prompt(prompt):
    client = openai.OpenAI(api_key=api_key)
    response = client.chat.completions.create(
        model="gpt-4",
        messages=[{"role": "user", "content": prompt}],
        max_tokens=2000
    )
    return response.choices[0].message.content
```

Model: OpenAI GPT-4 (API) as the base model.

Specialization: Fine-tuned (in later phases) on curated career coaching conversations, job search Q&A, and industry-specific advice datasets.

Prompt 1: Personalized Strengths Assessment

In [ ]:
```python
prompt = f"""
Given the following professional background, identify the user’s top strengths and
suggest how they can be leveraged in a job search.
Background:
{resume_text}
"""

print(openai_prompt(prompt))
```

```
Top Strengths:
1. Accomplished in data analysis and visualization: Pankaj Yadav has 15+ years of progressive experience in data transf
ormation, analysis, visualization, and process optimization. He has shown effectiveness in conducting cost-benefit anal
yses, scoring metrics, and identifying data trends and patterns.

2. Leadership and project management: Pankaj has demonstrated leadership experience, having led technical teams and spe
arheaded initiatives to improve efficiency, streamline reporting, and reduce operational risk. His experience in managi
ng complex data projects shows strategizing and planning abilities.

3. Technical proficiency: Pankaj has advanced proficiency in several programming languages and tools, including SAS, SQ
L, Python, Tableau, and various Microsoft applications. This includes specialized knowledge in seismic data processing
and modeling.

4. Industry knowledge in financial services and insurance: Pankaj has worked within the financial services and insuranc
e industries, making him highly knowledgeable about industry-specific data sets, regulations, and reporting needs.

5. Communication and collaboration: He has effectively collaborated cross-functionally to gather business requirements
and translated technical plans to stakeholders.

Leveraging these strengths in a job search:

1. Pankaj should emphasize his proficiency in data analysis and visualization in his resume and cover letters, highligh
ting examples where he has used these skills to influence business decisions.

2. He can point out his leadership and project management skills during interviews, sharing specific instances where he
has led teams to achieve project objectives.

3. His technical proficiencies should be underlined in the skills section of his resume and LinkedIn profile. It may al
so be beneficial to be endorsed on LinkedIn for these skills.

4. His industry knowledge can make him an asset for other companies in the same industries. Thus, during networking, he
can emphasize his deep understanding of the financial services and insurance sectors.

5. Pankaj should cite examples of how his collaborative approach and communication skills resulted in the success of cr
oss-functional projects, demonstrating how he can bring these same strengths to potential employers.
```

Prompt 2: Skill Gap Analysis and Development Plan

In [37]:

```python
prompt = f"""
Based on the user's resume and a target job description, identify key skill gaps and recommend a
step-by-step plan to close those gaps, including online courses or certifications.
Resume:
{resume_text}
```

```
Job Description:
{job_description}
"""

print(openai_prompt(prompt))
```

```
Skill Gaps:

1. Deep understanding of BI tools - The candidate has experience with Tableau but the job needs extensive expertise in enterprise BI tools such as Looker, Spotfire, Power BI.
2. AI-enabled solutions and Generative AI - The job description requires someone who can architect and lead the integration of AI-enabled solutions, particularly Generative AI.
3. Cloud Computing concepts - The job description mentions Google Cloud Platforms and an overall understanding of cloud computing concepts.
4. Data Pipelines - Experience with building and maintaining moderately-complex data pipelines.
5. Risk Modeling – The preferred qualification mentions background experience in risk modeling which is missing from the candidate's resume.
6. Advanced Git usage and CI/CD integration skills – These skills have not been mentioned in the resume.

Plan to Close Gaps:

1. Master BI Tools - The candidate should take online courses on enterprise BI tools. For instance, 'Data Visualization with Tableau' course on Coursera or 'Power BI A-Z: Hands-On Power BI Training for Data Science!' on Udemy can deepen their knowledge on BI tools.
2. AI-Enabled Solutions - For AI and Generative AI, AI for Everyone by Andrew Ng on Coursera is a good place to start. For more detailed understanding, AI Programming with Python Nano-degree on Udacity can be taken.
3. Cloud Computing Concepts - To understand Cloud Computing, the candidate can enroll in 'Google Cloud Platform Fundamentals: Core Infrastructure' on Coursera or 'AWS Certified Solutions Architect - Associate 2020 - Pass the Exam!' on Udemy for AWS cloud concepts.
4. Learn to Build Data Pipelines - 'Data Engineering, Big Data, and Machine Learning on GCP' and 'Designing and Building Data Warehouses' both on Coursera can enhance the candidate's skills in building and maintaining data pipelines.
5. Risk Modeling - 'Risk Management in Banking and Financial Markets Professional Certificate' on EdX can provide the candidate with in-depth knowledge of risk modeling.
6. Git and CI/CD - For Git and CI/CD, candidate can refer to 'Version Control with Git' and 'Continuous Integration' on Coursera.
7. Certifications - Lastly, corresponding certifications like 'Tableau Desktop Specialist', 'Google Cloud Certified - Professional Data Engineer' can be pursued to validate acquired skills.

Besides online courses, the candidate should also get hands-on by working on related projects or internships wherever possible. Peer-to-Peer learning platforms can also be beneficial.
```

Prompt 3: Industry-Specific Job Search Strategy

In [38]:
```
prompt = f"""
Based on the user’s resume, what are the most effective job
search strategies for someone seeking a role in Insurance?
Resume:
{resume_text}
"""

print(openai_prompt(prompt))
```

```
1. Networking: This can involve reaching out directly to potential employers, attending industry events, and trying to
develop relationships with people working in the insurance field.

2. Professional Social Networking Sites: LinkedIn can be a strong platform for job searching in the insurance field. Pa
rticipate in discussions, join groups related to the insurance industry, and follow companies that interest you.

3. Job Boards and Career Websites: Apart from general job boards, some sites are dedicated to the insurance industry. R
egularly checking and applying on these platforms can lead to potential job opportunities.

4. Recruiters and Headhunters: Given the candidate's extensive experience in the field, engaging with recruiters who sp
ecialize in insurance can help pinpoint job opportunities that may not always be publicly listed.

5. Tailored Applications: Each job application should be tailored to reflect the skills and experiences that are most r
elevant to the job in question. Given the candidate's technical skills, focusing on roles that require these skills wou
ld be wise.

6. Career Fairs: These events are often organized by universities, professional bodies, and companies to advertise job
opportunities and meet potential candidates.

7. Leveraging Professional Memberships: If the candidate is a member of any professional organizations (e.g. related to
data science or insurance), they should fully utilize the resources available, from networking events to job postings.

8. Keeping Up with Industry Trends: The candidate should closely follow the insurance industry, including trends and ne
ws. This information can come in handy during interviews and can also assist in identifying potential employers.

9. Informational Interviews: Reach out to professionals in targeted companies or roles to gain deeper insights about da
ily operations and key responsibilities. This can provide an edge while applying or preparing for interviews.

10. Continued Learning and Skills Development: The candidate should consider gaining additional certifications or learn
ing new tools and techniques that are common in the insurance industry. They should then highlight these on their resum
e, LinkedIn profile, and cover letter.
```

Prompt 4: Interview Preparation Guidance

In [39]:

```
prompt = f"""
What are the top 5 behavioral and technical interview questions for a data scientists in Banking industry,
and how should the user prepare for them?

Job Title:
Data Scientist

Industry:
Banking

Resume:
{resume_text}

Job Description:
{job_description}
"""

print(openai_prompt(prompt))
```

Behavioral Interview Questions:

1. Tell me about a time when your insights from the data analysis substantially impacted business decisions. What was your role in this, and what was the outcome?
   - To prepare for this question, analyze past projects where you contributed significantly to the decision-making process using the insights you gathered from your data analysis. Use specific examples and figures to demonstrate impact.

2. How have you collaborated with cross-functional teams in the past and can you provide an example where your input influenced the project's success?
   - Preparation should include outline of your experience working in multidisciplinary teams, your communication techniques, and how you approached conflict resolution. Provide a constructive example of how your input contributed to a project's success.

3. Can you tell me about a time when you faced significant opposition or skepticism about your data findings? How did you address this?
   - Prepare for this question by reflecting on instances where your data findings were challenged or disputed. Explain your approach to addressing skepticism, how you reinforced your findings, and the final outcome.

4. Can you describe a difficult problem you faced while working with data, and how you solved it?
   - Demonstrate your problem-solving skills and resilience by recounting a specific situation where you experienced difficulties while working with data, and how you solved it. Explain the actions you took, your thought process, and the results achieved.

5. Tell me about a time when you had to explain difficult technical concepts to a non-technical audience. How did you make sure they understood?
   - Reflect on past instances where you have had to simplify complex data or results to a non-technical audience. Discuss the techniques you used, and how you gauged their understanding.

Technical Interview Questions:

1. What is your experience with enterprise BI tools like Looker, Spotfire, Tableau, or Power BI? How do you handle their limitations?
   - You might want to review the BI tools you have used in the past, focusing on your accomplishments using them. Also, discuss any common limitations of these tools and how you worked around them.

2. Can you discuss a project where you had to extract, clean, and analyze a large data set using SQL and Python?
   - When preparing for this question, consider the steps you took including data extraction, cleaning, and analysis in SQL and Python. Highlight any challenges you faced and how you overcame them.

3. Tell me about your expérience in optimizing data models for high performance across massive data sets.
   - Describe your expertise in developing and optimizing data models. Discuss how you improved the efficiency of data retrieval and processing across large data sets.

4. How familiar are you with Google Cloud Platforms or other cloud computing concepts?

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   - Review your understanding and experience with Google Cloud Platforms, as well as other cloud environments. Give ex
amples of projects or tasks where you've applied these skills.

5. How have you used Git for version control in your projects? Can you describe a time when you used continuous integra
tion/ continuous delivery in data science projects?
   - Discuss your familiarity with Git and CI/CD by sharing experiences where you used these tools for version control
and automated software deployment respectively.
```

Prompt 5: Career Path Exploration

In [40]:

```python
prompt = f"""
Given the user’s current experience and interests, suggest three potential career paths,
including the pros and cons of each and the steps needed to transition.
User’s Experience and Interests based on Resume:
{resume_text}

Job Description:
{job_description}
"""

print(openai_prompt(prompt))
```

Potential Career Paths:

1. Business Intelligence Lead:

Pros: The role of a BI Lead will take advantage of Pankaj's strong analytics background and technical skills. As he has extensive work experience in data transformation, analysis visualization and process optimization, these skills will be critical in this role, especially in terms of architecting and scaling an industry-leading visualization function.

Cons: This role is mandate leading a team, thus if he’s not comfortable with leadership or lacks experience in managing a team, this could be a challenging aspect of the job. Also, the role requires high-level decision making which could mean high-stress situations.

Steps for Transition: Pankaj might want to gain more experience in a leading role, possibly seek additional training in management, and brush up on the knowledge of current best practices for data access, integration, and visualization within next-generation big data analytical platforms.

2. Data Science Consultant:

Pros: In this role, Pankaj’s broad ranged experience coupled with his Master of Science in Data Science could see him providing valuable insights and recommendations to clients on to improving their business operations. It's a role that can also offer a lot of variety and the opportunity to work on different projects with different clients.

Cons: A potential downside is the irregularity in job projects that might not provide the job security of a conventional role. The demands and expectations of different clients might also prove stressful.

Steps for Transition: A good step would be to network with professionals in consulting to understand the ins and outs of the role. He’d potentially need to cultivate a strong personal brand to attract clients and projects.

3. Chief Data Officer:

Pros: Given his extensive experience within the field, Pankaj might consider a role as a Chief Data Officer in organizations where their focus is using data to drive decision-making and operations. In this role, he could potentially lead strategic decisions for the company's data handling.

Cons: It is a high-pressure and high-responsibility role that needs a thorough understanding of business strategy in addition to technical expertise. The CDO role often involves much more corporate policy and strategic decision making which may be different from his previous roles.

Steps for Transition: Pankaj might want to seek further education/training in business strategies and corporate decision making. A more varied role in data management would provide a wider experience base to draw from. Networking with other professionals in similar roles might also be beneficial.